

Advanced React Masterclass – Module 2 Exercise

Hands-on Exercise No. 2

Course: Advanced React Masterclass

Module: Hooks & Context

Total Marks: 10

Due Date: 14/05/2026

Instructions:

- Prepare your solution in MS Word
 - Submit in .doc or .docx format
 - Late or copied submissions will not be accepted
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Learning Outcome:

- Use useState and useEffect hooks effectively
 - Create reusable custom hooks
 - Implement Context API for global state
 - Manage complex state using useReducer
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Problem Statement:

You are working as a Frontend Engineer for a SaaS company. The client requires a Theme Switcher application to demonstrate React hooks and global state management using Context API.

Task Requirements:

- Create a new React project
- Implement a Theme Context using Context API
- Use useState to toggle between Light Mode and Dark Mode
- Use useEffect to log theme changes in the console
- Create a custom hook (useTheme) to manage theme logic
- Use useReducer to handle multiple theme states (Light, Dark, High Contrast)

- Display current theme on screen and allow switching using buttons
 - Take screenshots of:
 - Context setup
 - Custom hook implementation
 - Browser output showing theme switching
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Submission Guidelines:

- Upload your .docx file on LMS
- Include:
 - Code snippets
 - Screenshots of application
 - Output results
- Add short explanations for each feature